

RBE211TC

Dynamic Systems
School of Robotics Stage 3 | Level 2

SECTION A: Basic Information

Brief Introduction to the Module

RBE211TC (Dynamic Systems) is a 2.5-credit module offered by the School of Robotics. The module introduces students to the mathematical modelling and analysis of dynamic systems commonly encountered in robotic and mechatronic applications. Students will gain an understanding of system behaviour in the time domain, as well as fundamental stability characteristics relevant to motion control and system response.

The module integrates theoretical instruction with laboratory-based activities to support the modelling, analysis, and interpretation of dynamic system behaviour through experimental investigation. In addition to a closed-book examination, students are required to complete a laboratory portfolio.

Key Module Information

Module name	Module code	Credit value	Semester in which the module is taught	Pre-requisites needed for the module
Dynamic Systems	RBE211TC	2.5	SEM2	RBE101TC AND MTH007
Programmes on which the module is shared	BEng Intelligent Robotics Engineering with Contemporary Entrepreneurialism			

Module Leader and Contact Details

Module Leader:

Name	Email address	Office telephone number	Room number	Office hours	Preferred means of contact
Sze-Hong Teh	Szhong.Teh@xjtlu.edu.cn	88970752	A-4034	14:00-16:00 Wednesday	Email
Brief Biography	Dr. Sze-Hong Teh is an Assistant Professor in the School of Robotics at Xi'an Jiaotong-Liverpool University (XJTLU). He earned his MEng (Hons) in Mechanical Engineering in 2011 and his PhD in Mechanical Engineering in 2016, both from the University of Nottingham, where he later taught and conducted research before joining XJTLU in 2021. He is an associate member of the UK Institution of Mechanical Engineers (AMIMechE) and a fellow of the UK Higher Education Academy (FHEA). His research focuses on applied nonlinear dynamics, biomimetic robotics, intelligent control systems, computer vision, and machine learning.				

Additional Teaching Staff and Contact Details:

Role	Name	Email address	Office telephone number	Room number	Office hours	Preferred means of contact
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SECTION B: What You Can Expect from the Module

Educational Aims of the Module

The Dynamic Systems module introduces the fundamental principles of mathematical modelling and analysis of dynamic systems to support the understanding of system behaviour in the time domain as well as system stability characteristics, which are important in the design of robotic and mechatronic systems.

The module aims to:

1. develop an understanding of modelling methods for mechanical systems;
2. introduce the fundamental principles governing the behaviour of first- and second-order dynamic systems in the time domain; and
3. develop practical skills in designing mechatronic systems through conducting and reporting simple experiments.

Learning Outcomes

Students completing the module should be able to:

- A. Model dynamic systems and analyse their characteristics across mechanical, thermal, fluid, and electrical domains, demonstrating a comprehensive understanding.
- B. Apply essential principles governing the behavior of first and second order systems in the time domains, showcasing proficiency in practical applications.
- C. Apply advanced analytical methods to determine natural frequencies and mode shapes and understand the stability issues in mechanical systems.
- D. Design and optimise simple mechatronic systems to achieve the specified objectives related to performance, cost, safety, environmental protection and sustainability.

Methods of Learning and Teaching

The teaching philosophy of the module is aligned with the principles of Syntegrative Education. The module integrates theoretical instruction with laboratory-based learning to support students in understanding the modelling and analysis of dynamic systems relevant to robotic and mechatronic applications.

This module is delivered through a combination of formal lectures, supervised laboratory sessions, and a review seminar. Lectures introduce the theoretical foundations of dynamic system modelling and analysis, supported by worked examples. Supervised laboratory sessions provide opportunities for students to conduct experiments on representative mechanical and electromechanical systems, and to develop practical skills in analysing dynamic system responses through experimental investigation.

The module is assessed through a combination of a laboratory-based portfolio and a closed-book examination.

Syllabus & Teaching Plan

Week Number	Mode of Delivery (Lecture/Tutorial/Seminar/Field Trip/Other)	Topic	Pre-reading and others
W1	Lecture/Practical	Lecture 1 & 2 • Introduction to Dynamic Systems • Modelling Mechanical Systems Lab 1 • Free Vibration of a Mass-Spring System	• Lecture Notes 1 • Lab Sheet 1

W2	Lecture/Practical	Lecture 3 & 4 - Modelling Electrical and Electromechanical Systems Lab 2 - Free Vibration of a Simple Pendulum System	- Lecture Notes 2 - Lab Sheet 2
W3	Lecture/Practical	Lecture 5 & 6 - Lagrangian Method - Analytical Solution of Linear Dynamic Systems Lab 3 - Experimental Estimation of Damping Ratio	- Lecture Notes 3 - Lab Sheet 3
W4	Lecture/Practical	Lecture 7 & 8 - Vibration of Mechanical Systems Lab 4 - Electrical RLC System	- Lecture Notes 4 - Lab Sheet 4
W5	Lecture/Practical	Lecture 9 & 10 - Vibration of Mechanical Systems Lab 5 - Multi-DOF Rotary Torsional System	- Lecture Notes 5 - Lab Sheet 5
W6	Lecture/Practical/Seminar	Lecture 11 & 12 - Dynamic Systems in Feedback Control Lab 6 - Overhead Crane Anti-Sway System Seminar - Revision	- Lecture Notes 6 - Lab Sheet 6

Assessment Details

Initial Assessment

Final Exam (55% of the module mark)

Assessment Type: EXAM

Learning outcomes assessed: ABC

Duration: 2.0 hours

Resit opportunity: Y

Assessment Task	Learning Outcomes	Weighting	Release Date	Due Date
Final Exam	ABC	55%	/	/
Artificial Intelligence Permissions	/			
Brief Description of the Assessment Task				

1. This is a closed-book examination, which is to be written without books or notes.
2. Total marks available are 90 and the final mark will be scaled to 100%.
3. This paper consists of three questions. Answer ALL questions.
4. Only silent, non-programmable calculators are permitted in this examination.
5. Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.
6. Answer for each question should start on a NEW page in answer booklet.
7. Please write down your solution in answer booklet with a clear and neat handwriting.
8. Please be reminded that correct answers do not guarantee a full mark: mark penalty will be imposed for missing intermediate steps, illogical process, poor presentation, missing relevant illustration, erroneous notation, etc.
9. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Laboratory (45% of the module mark)

Assessment Type: CW

Learning outcomes assessed: ABCD

Duration: N/A

Resit opportunity: N

Assessment Task	Learning Outcomes	Weighting	Release Date	Due Date
Laboratory	ABCD	45%	02/Mar/2026	19/Apr/2026 23:59
Artificial Intelligence Permissions	The use of Generative AI is not permitted.			

Brief Description of the Assessment Task

Students are required to complete a laboratory data sheet that documents experimental data, calculations, plots, and brief reflections from a series of laboratory activities. The data sheet will be progressively completed throughout the semester and submitted as a single PDF file at the end of the module.

The submitted data sheet should contain entries for all laboratory sessions attended. Students are expected to attend all laboratory sessions. Absence from laboratory sessions is likely to result in incomplete experimental records, which will be reflected in reduced performance in the Laboratory Data Sheet and corresponding marks.

Resit Assessment

Exam (55% of the module mark)

Assessment Type: EXAM

Learning outcomes assessed: ABC

Duration: 2.0 hours

Assessment Task	Learning Outcomes	Weighting	Release Date	Due Date
Exam	ABC	55%	/	/
Artificial Intelligence Permissions	/			

Brief Description of the Assessment Task

1. This is a closed-book examination, which is to be written without books or notes.
2. Total marks available are 90 and the final mark will be scaled to 100%.
3. This paper consists of three questions. Answer ALL questions.
4. Only silent, non-programmable calculators are permitted in this examination.
5. Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.
6. Answer for each question should start on a NEW page in answer booklet.
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Reading Materials

Type	Title	Author	ISBN/Publisher									
Mandatory Textbooks	N/A											
Optional Textbooks	N/A											
Reference Textbooks	SYSTEM DYNAMICS MODELING, SIMULATION, AND CONTROL OF MECHATRONIC SYSTEMS, 5TH EDITION	DEAN C. KARNOPP, DONALD L. MARGOLIS, RONALD C. ROSENBERG	9780470889084/WILEY (FEBRUARY 28, 2012)									
	PROCESS DYNAMICS MODELING, ANALYSIS AND SIMULATION	B. WAYNE BEQUETTE	9780132068895/PRENTICE HALL INTERNATIONAL, 1998									
	SYSTEM DYNAMICS: AN INTRODUCTION, 1 EDITION	DEREK ROWELL, DAVID N. WARMLEY	9780132108089/PEARSON (OCTOBER 10, 1996)									
Additional Materials	<table><tr><td>MECHANICAL VIBRATIONS, 5TH EDITION</td><td>SINGIRE SU S. RAO</td><td>9781292178608 /PEARSON (2017)</td></tr><tr><td>MODERN CONTROL SYSTEMS, 13TH EDITION</td><td>DORF BISHOP</td><td>9780134407623 /PEARSON (2016)</td></tr><tr><td>DYNAMIC SYSTEMS - MODELING, SIMULATION, AND CONTROL, 2ND EDITION</td><td>CRAIG A. KLUEVER</td><td>9781119723479 /WILEY(2020)</td></tr></table>			MECHANICAL VIBRATIONS, 5TH EDITION	SINGIRE SU S. RAO	9781292178608 /PEARSON (2017)	MODERN CONTROL SYSTEMS, 13TH EDITION	DORF BISHOP	9780134407623 /PEARSON (2016)	DYNAMIC SYSTEMS - MODELING, SIMULATION, AND CONTROL, 2ND EDITION	CRAIG A. KLUEVER	9781119723479 /WILEY(2020)
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SECTION C: Additional Information

This section provides students with essential information and resources pertaining to their academic studies to ensure a successful academic journey and engagement with the module.

Student Feedback:

The University is committed to receiving and responding to student feedback in order to improve the quality of the student experience within the institution. It is University policy that the preferred way of doing this is by using the Online Student Module Feedback Questionnaire Survey. Students are encouraged to complete the questionnaire survey for this module at the end of the semester.

Attendance:

The University expects students to attend all timetabled learning sessions associated with this module, and to engage with the relevant learning and support resources. Student attendance will be recorded using the Attendance Management System (AMS). Please follow your teacher's instructions for recording your attendance at each session. Students are responsible for managing their attendance, and should take prompt action to inform the Module Leader in case circumstances beyond their control affect their class attendance. You are advised to read the University's 'Student Attendance Policy' for more information.

Rules of Submission for Assessed Coursework:

The University has detailed rules and procedures governing the submission of assessed coursework. You need to be familiar with the rules and procedures as detailed in the University's 'Code of Practice on Assessment'.

Late Submission of Assessed Coursework:

The University attaches penalties to the late submission of assessed coursework. You need to be familiar with the rules as detailed in the University's 'Code of Practice on Assessment'.

Mitigating Circumstances:

Students who are unable to submit coursework by the deadline or attend examinations due to serious illness or other unforeseen circumstances, as defined in the Mitigating Circumstances (MC) Policy, may submit a Mitigating Circumstances Application. The application should be submitted before the assessment deadline (or examination date) under the Academic Records page on e-Bridge. Misuse of the MC policy will result in disciplinary actions and demerit points.

Academic Integrity:

Any violation of academic integrity including plagiarism, collusion, copying, submission of commissioned or procured work, Academic Offences Involving Artificial Intelligence (AI), and/or falsification and fabrication of data will result in penalties and demerit points. Please be familiar with the University's Academic Integrity Policy.

Examination Misconduct:

The University also values academic integrity in the conduct of examinations. Any behavior that violates examination regulations will not be tolerated and will result in penalties.

Artificial Intelligence (AI):

Information on whether the use of AI is permitted or not for each assessed coursework is indicated in the Assessment Details section of this module handbook.

For detailed guidelines, please refer to the Academic Integrity Policy Guide for Students and Staff.

For more information and resources on Artificial Intelligence and your learning and assessment, please consult the XJTLU AI for Learning pages of the Learning Mall Core.

Learning Mall Core:

Copies of lecture notes and other materials are available electronically through the Learning Mall Core, the University's virtual learning environment, at learningmall@xjtlu.edu.cn.

Communication:

All official communication concerning module-related matters will be conducted via e-mail and/or as Learning Mall Core announcements. Other modes of electronic communication are treated as informal.

Further Support:

You are advised to contact your Module Leader in the first instance if you experience any issues with your learning on this module. You may also contact your Academic Advisor or Programme Director. Further information on the kinds of support that the University provides to students can be found in the XJTLU Student Handbook.

You are strongly advised to read the policies mentioned above very carefully, because this will help you perform better in your academic studies. You can find all the policies and regulations related to your academic study on the e-Bridge → 'Document Zone' page.

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Cut-off Date: 28/Feb/2026